

## Frozen inputs

ESM-2 650M layer 18  
278 discovery proteins  
InterPLM SAE  
 $1280 \times 10240$

## Stage-B probe directions

L1 logistic,  $N=139$   
200 canonical fits  
byte-verified replay

## Decoder geometry

OMP,  $k$  grid 1...512  
joint LS refit  
 $w = D^T \beta$

## Descriptive summary

median  $R(32)$ , top-32  
mass,  $N_{\text{eff}}$   
10,000 bootstrap

*four frozen comparison populations*

## Biological

100 / 100 eligible

## Canonical null

63 / 100 eligible

## C-matched null

100 / 100 eligible

## Isotropic reference

1000 directions